

```

1: ////////////////////////////////////////////////// Test 2, 2023/2024 (Sem 1) ///////////////////////////////////
2: //// QUESTION 2
3: //// Name: Nurul Ika Syafiny Binti Azhar
4: //// Matric No: A23CS0164
5: //// Date / Day: 4/1/2024 Thursday
6: ///////////////////////////////////
7:
8: #include<iostream>
9: #include<iomanip>
10: using namespace std;
11:
12: void displayMenu();           // Task 6. Function prototypes
13: void getUserInput(int *);
14: float calculateLateFees(int );
15: void displayLibraryInfo();
16: // Task 5. Main function
17: int main()
18: {
19:     int menuoption, daysoverdue = 0;
20:     float latefees;
21:
22:     do
23:     {
24:         displayMenu();
25:         cout<<"Enter your choice(1-3): ";
26:         cin>>menuoption;
27:
28:         switch( menuoption)
29:         {
30:             case 1:
31:                 getUserInput(&daysoverdue);
32:                 latefees = calculateLateFees(daysoverdue);
33:                 cout<<"Late fees: RM"<<fixed<<setprecision(2)<<latefees<<endl;
34:                 break;

```

```

35:
36:         case 2:
37:             displayLibraryInfo();
38:             break;
39:
40:         case 3:
41:             break;
42:
43:         default:
44:             cout<<"Invalid choice. Please enter a number between 1 and 3\n";
45:     }
46: }while(menuoption != 3);
47:
48: cout<<"\nExiting the program. Thankyou!";
49:
50: return 0;
51: }
52:
53: // Task 2. Function displayMenu
54: void displayMenu()
55: {
56:     cout<<"*****Library Book Checkout System*****"<<endl;
57:     cout<<"1.Calculate Late Fees\n2.Display Library Information\n3.Quit"<<endl;
58: }
59: // Task 4. Function getUserInput with reference parameter
60: void getUserInput(int *daysoverdue)
61: {
62:     cout<<"\nEnter the number of days the book is overdue: ";
63:     cin>>*daysoverdue;
64: }
65: // Task 1. Function calculateLateFees
66: float calculateLateFees(int daysoverdue)
67: {
68:     float totalfees;

```

```
69:
70:     if(daysoverdue<=7)
71:     {
72:         totalfees = 0.5*daysoverdue;
73:     }
74:     else if((daysoverdue>7) && (daysoverdue<=14))
75:     {
76:         totalfees = 0.5*7.0 + (daysoverdue - 7);
77:     }
78:     else
79:     {
80:         totalfees = 0.5*7.0 + 7.0 + (daysoverdue - 14);
81:     }
82:     return totalfees;
83: }
84: // Task 3. Function displayLibraryInfo
85: void displayLibraryInfo()
86: {
87:     cout<<"Library Name: UTM Library\nAddress: Skudai, Johor\nContact: (123) 456 7890\n";
88: }
89: // Task 7. Directives, library, header, etc.
90:
91:
92:
93:
94:
95:
96:
97:
98:
99:
100:
101:
102:
```

103:
104:
105:
106:
107:
108:
109: